VITAGEMINA

Clinical AI Twin System · v0.1.0

Optional local AI offline · Twin active 22:50:03 · Dario Arian · #PT-00472

⚠ **Organ index below threshold:** Hepatic 70.2/75 · Neuro 60.3/70 · Metabolic 55.6/70

Operational context · Browser demo · Client-side simulation · Optional local AI · Production requires secure hosting, FHIR/HL7 integration, SSO/MFA

CARDIAC MONITOR

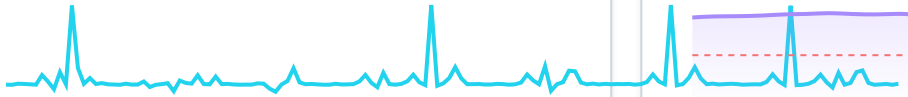
SINUS RHYTHM

Heart rate

78 BPM

SpO_2

98 %



PQRST morphology · Lead II

CONTINUOUS GLUCOSE

IN RANGE

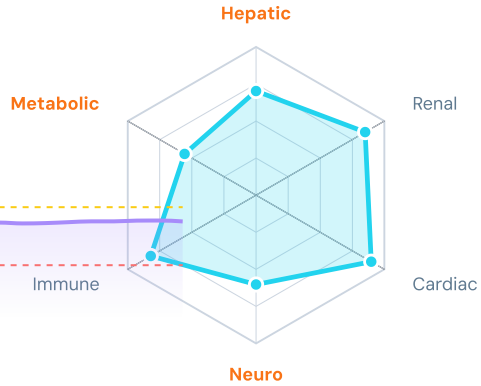
Current glucose

122.8 mg/dL

Target 70–140

ORGAN SYSTEM INDEX

6-domain · /100



Hepatic	70.2
Renal	85
Cardiac	89.7
Neuro	60.3
Immune	82
Metabolic	55.6

BLOOD CHEMISTRY PANEL

WBC	HbA1c	Creatinine
6.8	7.1	0.92
K/µL	%	mg/dL
	→ Borderline	

PHARMACOGENOMIC PROFILE

CYP2D6	Accelerated drug clearance via CYP2D6 pathway	UM
BRCA2	No elevated hereditary cancer risk	WT

LDL-C	CRP	eGFR
143 mg/dL → Borderline	3.8 mg/L ↑ Elevated	84 mL/min

TPMT	Moderate thiopurine sensitivity — reduce dose	IM
DPYD	Fluoropyrimidine (5-FU) toxicity — critical contraindication	PM
VKORC1	Reduced warfarin dose requirement	VAR
HLA-B	No abacavir / carbamazepine hypersensitivity	WT

DRUG SIMULATION ENGINE

Carboplatin AUC5 | Reset

Efficacy	Toxicity risk	Half-life	Optimal dose
79%	22%	2.4h	85%

Plasma concentration · µg/mL · 0–12h

--- Population average Dario Arian (genomic)

PREDICTIVE EVENT TIMELINE

Arrhythmia risk	~4.2h	16.6% Low risk
Glucose dip <70	~1.8h	39.5% Moderate risk
CRP spike	~7.0h	60.2% High probability

Ensemble model: CGM trend · ECG pattern · CRP trajectory · pharmacogenomic weighting. Probabilities refresh every 3 s in active mode.

Biomed-Ensemble-v3 · Twin 4.2.1-dario · Sync: live-simulated · Optional local AI explanation

> TPMT intermediate metabolizer status predicts extended exposure and elevated myelosuppression risk.

TPMT IM detected — recommend 15% dose reduction and enhanced CBC monitoring on days 7 and 14.